



1. Data Preparation

a. Import the data into a DataFrame

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("sales.csv")
df.head()
```

```
Out[1]:
```

	ID	Product	Category	Quantity	PricePerUnit	Date
0	1	Smartphone	Electronics	10	25.50	1/1/2024
1	2	Laptop	Electronics	8	18.75	1/2/2024
2	3	Headphones	Electronics	15	12.90	1/3/2024
3	4	Refrigerator	Appliances	12	30.00	2/1/2024
4	5	Blender	Appliances	5	22.50	2/2/2024

b. Adding new Cloumn

```
In [7]: df["Total_Sale"] = df["Quantity"] * df["PricePerUnit"]
df.head()
```

```
Out[7]:
```

	ID	Product	Category	Quantity	PricePerUnit	Date	Total_Sale
0	1	Smartphone	Electronics	10	25.50	1/1/2024	255.0
1	2	Laptop	Electronics	8	18.75	1/2/2024	150.0
2	3	Headphones	Electronics	15	12.90	1/3/2024	193.5
3	4	Refrigerator	Appliances	12	30.00	2/1/2024	360.0
4	5	Blender	Appliances	5	22.50	2/2/2024	112.5

c. Mean, Median, and Standard Deviation

```
In [9]: mean_sale = df["Total_Sale"].mean()
median_sale = df["Total_Sale"].median()
std_sale = df["Total_Sale"].std()

print("Mean Total Sale: ", mean_sale)
print("Median Total Sale: ", median_sale)
print("Standard Deviation of Total Sale: ", std_sale);
```

Mean Total Sale: 325.422
Median Total Sale: 300.0
Standard Deviation of Total Sale: 133.83027970779509

2. Visualization

a. Bar graph showing Total_sale by month

```
In [40]: df["Date"] = pd.to_datetime(df["Date"])  
df.head()
```

```
Out[40]:
```

	ID	Product	Category	Quantity	PricePerUnit	Date
0	1	Smartphone	Electronics	10	25.50	2024-01-01
1	2	Laptop	Electronics	8	18.75	2024-01-02
2	3	Headphones	Electronics	15	12.90	2024-01-03
3	4	Refrigerator	Appliances	12	30.00	2024-02-01
4	5	Blender	Appliances	5	22.50	2024-02-02

```
In [41]: df["Month"] = df["Date"].dt.month  
df.head()
```

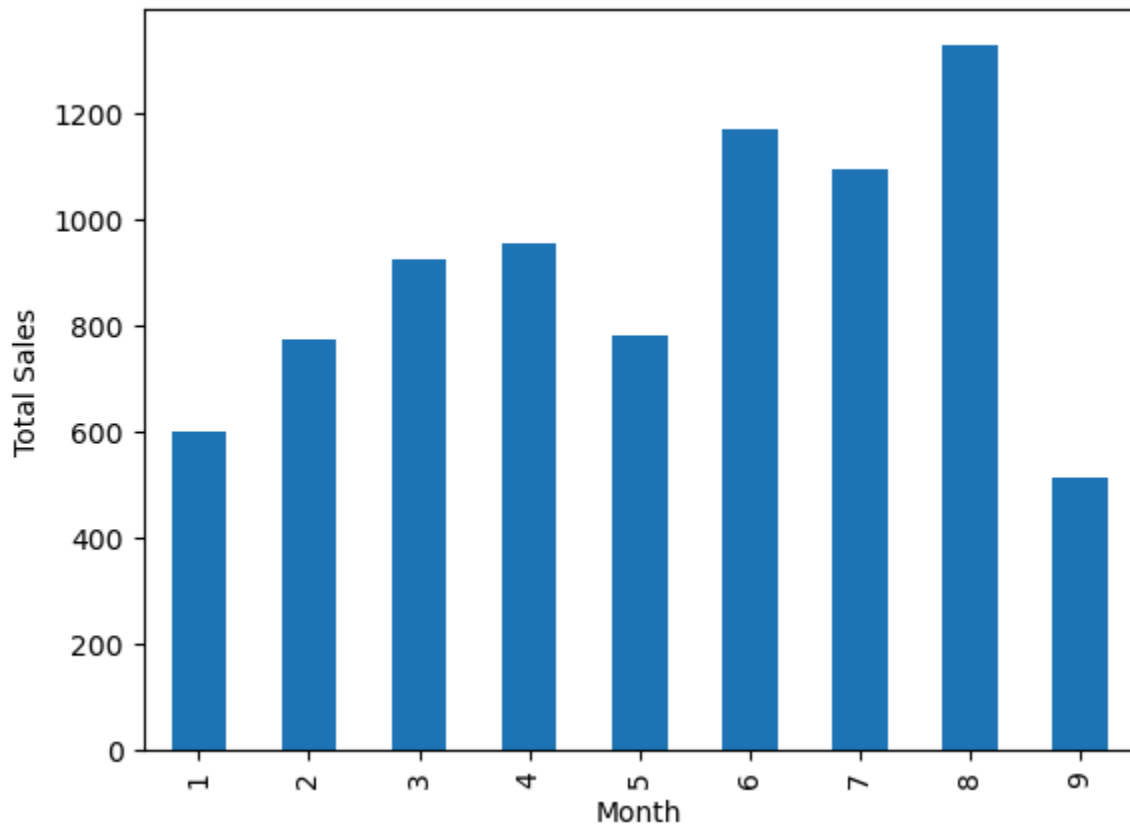
```
Out[41]:
```

	ID	Product	Category	Quantity	PricePerUnit	Date	Month
0	1	Smartphone	Electronics	10	25.50	2024-01-01	1
1	2	Laptop	Electronics	8	18.75	2024-01-02	1
2	3	Headphones	Electronics	15	12.90	2024-01-03	1
3	4	Refrigerator	Appliances	12	30.00	2024-02-01	2
4	5	Blender	Appliances	5	22.50	2024-02-02	2

```
In [16]: monthly_sales = df.groupby("Month")["Total_Sale"].sum()  
monthly_sales
```

```
Out[16]: Month  
1      598.50  
2      772.50  
3      925.00  
4      952.50  
5      782.55  
6     1169.00  
7     1095.00  
8     1328.50  
9      512.00  
Name: Total_Sale, dtype: float64
```

```
In [24]: plt.figure()
monthly_sales.plot(kind="bar")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.show()
```



b. Scatter plot: Quantity and Price Correlation

```
In [33]: plt.figure()
plt.scatter(df["Quantity"], df["PricePerUnit"])
plt.xlabel("Quantity")
plt.ylabel("Price Per Unit")
plt.title("Quantity vs Price Per Unit")
plt.show()

correlation = df["Quantity"].corr(df["PricePerUnit"])
print("Correlation between Quantity and Price: ", correlation)
```



Correlation between Quantity and Price: -0.5163718700038966

Findings:

The correlation between Quantity and Price Per Unit is -0.52 , which indicates a moderate negative correlation. This suggests that, in general, products sold in higher quantities tend to have lower prices, while higher-priced items are usually sold in smaller quantities. The scatter plot supports this observation, showing a downward trend as quantity increases. However, the relationship is not perfectly linear, implying that other factors may also influence sales volume.

3. Filtering & Export

a. Filter rows

```
In [43]: filtered_df = df[(df["Quantity"] > 10) & (df["Month"] > 5)]
         filtered_df
```

Out[43]:	ID	Product	Category	Quantity	PricePerUnit	Date	Month
15	16	Air Fryer	Appliances	13	32.00	2024-06-01	6
17	18	Vacuum Cleaner	Electronics	30	20.00	2024-06-03	6
19	20	Microwave	Appliances	12	25.00	2024-07-02	7
20	21	Drone	Electronics	28	18.75	2024-07-03	7
21	22	Camera	Electronics	14	35.00	2024-08-01	8
22	23	Iron	Electronics	11	28.50	2024-08-02	8
23	24	Soundbar	Appliances	35	15.00	2024-08-03	8
24	25	Laptop Stand	Appliances	16	32.00	2024-09-01	9

c. Save filtered data

```
In [44]: filtered_df.to_csv("filtered_sales.csv", index=False)
f_df = pd.read_csv("sales.csv")
f_df.head()
```

Out[44]:	ID	Product	Category	Quantity	PricePerUnit	Date
0	1	Smartphone	Electronics	10	25.50	1/1/2024
1	2	Laptop	Electronics	8	18.75	1/2/2024
2	3	Headphones	Electronics	15	12.90	1/3/2024
3	4	Refrigerator	Appliances	12	30.00	2/1/2024
4	5	Blender	Appliances	5	22.50	2/2/2024